Exercise 15A

Solution 1:

(i) Here, $l = 12$ cm, $b = 8$ cm, $h = 4.5$ cm

Volume of the cuboid $= l \times b \times h$

$= (12 \times 8 \times 4.5)\ cm^3$

$= 432\ cm^3$

Total Surface area $= 2(lb + lh + bh)$

$= 2(12 \times 8 + 12 \times 4.5 + 8 \times 4.5)\ cm^2$

$= 2(96 + 54 + 36)\ cm^2$

$= 2 \times 186\ cm^2$

$= 372\ cm^2$

Lateral surface area $= 2(l + b) \times h$

$= [2(12 + 8) \times 4.5]\ cm^2$

$= [2(20) \times 4.5]\ cm^2$

$= 40 \times 4.5\ cm^2$

$= 180\ cm^2$

(ii) Here, $l = 26$ m; $b = 14$ m; $h = 6.5$ m

Volume of the cuboid $= l \times b \times h$

$= (26 \times 14 \times 6.5)\ m^3$

$= 2366\ m^3$

Total surface area $= 2(lb + lh + bh)$

$= 2(26 \times 14 + 26 \times 6.5 + 6.5 \times 14)\ m^2$

$= 2(364 + 169 + 91)\ m^2$

$= 2 \times 624\ m^2$

$= 1248\ m^2$

Lateral surface area $= 2(l + b) \times h$

$= [2(26 + 14) \times 6.5]\ m^2$

$= [2 \times 40 \times 6.5]\ m^2$

$= 520\ m^2$

(iii) Here, $l = 15$ m; $b = 6$ m; $h = 5$ dm $= 0.5$ m

Volume of the cuboid $= l \times b \times h$

$= (15 \times 6 \times 0.5)\ m^3$

$= 45\ m^3$

Total surface area $= 2(lb + lh + bh)$

$= 2(15 \times 6 + 15 \times 0.5 + 6 \times 0.5)\ m^2$

$= 2(90 + 7.5 + 3)\ m^2$

$= 2 \times 100.5\ m^2$

$= 201 m^2$

Lateral surface area $= 2\ (l + b) \times h\ 2$

$= [2(15 + 6) \times 0.5]\ m^2$

$= [\ 2 \times 21 \times 0.5]\ m^2$

$= 21\ m^2$

(iv) Here, $l = 24$ m; $b = 25$ cm $= 0.25$ m; $h = 6$ m

Volume of the cuboid $= l \times b \times h$

$= (24 \times 0.25 \times 6)\ m^3$

$= 36\ m^3$

$$\text{Total Surface area} = 2(lb + lh + bh)$$
$$= 2(24 \times 0.25 + 24 \times 6 + 0.25 \times 6)\ m^2$$
$$= 2(6 + 144 + 1.5)\ m^2$$
$$= 2 \times 151.5\ m^2$$
$$= 303\ m^2$$

$$\text{Lateral surface area} = 2\ (l + b) \times h$$
$$= [2(24 + 0.25) \times 6]\ m^2$$
$$= [2 \times 24.25 \times 6]\ m^2$$
$$= 291\ m^2$$

Solution 2: For a matchbox,

Length $= 4$ cm

Breadth $= 2.5$ cm

Height $= 1.5$ cm

$$\text{Volume of each matchbox} = \text{Volume of cuboid}$$
$$= \text{Length} \times \text{Breadth} \times \text{Height}$$
$$= 4 \times 2.5 \times 1.5$$
$$= 15\ cm^3$$

$$\therefore \text{Volume of 12 matchboxes} = 12 \times 15$$
$$= 180\ cm^3$$

Thus, the volume of a packet containing 12 such matchboxes is 180 cm^3.

Solution 3: For a cuboidal water tank,

Length $= 6$ m

Breadth $= 5$ m

Height $= 4.5$ m

Volume of water in the tank = Length × Breadth × Height

$= 6 \times 5 \times 4.5$

$= 135\ m^3$

Now,

Volume of the cuboidal water tank = Length × Breadth × Height

$= (6 \times 5 \times 4.5)\ m^3$

$= 135\ m^3$

$= 135 \times 1000$ litres $\quad [\because 1\ m^3 = 1000$ litres$]$

$= 135000$ litres

Thus, the water tank can hold 135000 L of water.

Solution 4: For a cuboidal water tank,

Length $= 10$ m

Height (depth) $= 2.5$ m

Volume $= 50000$ litres $= \frac{50000}{1000} = 50\ m^3 \qquad (\because 1000\ L = 1\ m^3)$

Now,

Volume of a cuboidal tank = Length × Breadth × Height

$\Rightarrow 50 = 10 \times$ Breadth $\times 2.5$

$\Rightarrow$ Breadth $= \frac{50}{10 \times 2.5}$

$\Rightarrow$ Breadth $= \frac{50}{25}$

$\Rightarrow$ Breadth $= 2$ m

Thus, the depth (breadth) of a tank is 2 m.

Solution 5: Volume of the godown $= 40\ m \times 25\ m \times 15\ m = 15000\ m^3$

Volume of each wooden crate $= 1.5\ m \times 1.25\ m \times 0.5\ m = 0.9375\ m^3$

$\therefore$ Maximum number of wooden crates that can be stored in the godown

$= \frac{\text{Volume of the godown}}{\text{Volume of each wooden crate}}$

$$= \frac{15000}{0.9375}$$

$$= 16000$$

Solution 6:

$$\text{Number of planks} = \frac{\text{volume of the pit in cm}^3}{\text{volume of 1 plank in cm}^3}$$

$$\text{Volume of one plank} = (l \times b \times h)\ \text{cm}^3$$

$$= 500 \times 25 \times 10\ \text{cm}^3$$

$$= 125000\ \text{cm3}$$

Here, $l = 20$ m $= 2000$ cm; $b = 6$ m $= 600$ cm; $h = 80$ cm

$$\therefore \text{Volume of the pit} = (l \times b \times h)\ \text{cm}^3$$

$$= 2000 \times 600 \times 80\ \text{cm}^3$$

$$= 96000000\ \text{cm}^3$$

$$\therefore \text{Number of planks} = \frac{\text{Volume of the pit}}{\text{Volume of one plank}} = \frac{96000000}{125000} = \frac{96000}{125} = 768$$

Solution 7:

Length of the wall $= 8$ m $= 800$ cm

Breadth of the wall $= 22.5$ cm

Height of the wall $= 6$ m $= 600$ cm

i.e., volume of wall $= 800 \times 22.5 \times 600\ \text{cm}^3 = 10800000\ \text{cm}^3$

Length of the brick $= 25$ cm

Breadth of the brick $= 11.25$ cm

Height of the brick $= 6$ cm

i.e., volume of one brick $= 25 \times 11.25 \times 6 = 1687.5\ \text{cm}^3$

$$\therefore \text{ Number of bricks} = \frac{\text{volume of the wall}}{\text{volume of one brick}} = \frac{10800000}{1687.5} = 6400$$

Solution 8:

Length of the cistern, $l = 8$ m

Breadth of the cistern, $b = 6$ m

Height (or depth) of the cistern, $h = 2.5$ m

$\therefore$ Capacity of the cistern

$=$ Volume of the cistern

$= l \times b \times h$

$= 8 \times 6 \times 2.5$

$= 120\ \text{m}^3$

Also,

Area of the iron sheet required to make the cistern

$=$ Total surface area of the cistern

$= 2(lb + bh + hl)$

$= 2(8 \times 6 + 6 \times 2.5 + 2.5 \times 8)$

$= 2 \times 83$

$= 166\ \text{m}^2$

Solution 9:

Length of the room, $l = 9$ m

Breadth of the room, $b = 8$ m

Height of the room, $h = 6.5$ m

Area of four walls of the room $= 2(\text{length} + \text{breadth}) \times \text{Height}$

$= [2(9 + 8) \times 6.5]\ \text{m}^2$

$= (34 \times 6.5)\ m^2$

$= 221\ m^2$

Area of one door = Length × Breadth = $(2 \times 1.5)\ m^2 = 3\ m^2$

Area of two windows = 2 × (Length × Breadth)

$= [2 \times (1.5 \times 1)]\ m^2$

$= (2 \times 1.5)\ m^2$

$= 3\ m^2$

Area to be whitewashed

= Area of four walls of the room − Area of one door − Area of two windows

$= (221 - 3 - 3)\ m^2$

$= 215\ m^2$

Cost of whitewashing = ₹ 25 per square metre

⇒ Cost of whitewashing 215 m^2 = ₹ (25×215)

= ₹ 5375

Solution 10:

Length of the wall = 15 m = 1500 cm

Breadth of the wall = 30 cm

Height of the wall = 4 m = 400 cm

Volume of wall = $1500 \times 30 \times 400\ cm^3 = 18000000\ cm^3$

Now, volume of each brick = $22 \times 12.5 \times 7.5\ cm^3$

$= 2062.5\ cm^3$

Also, volume of the mortar = $\frac{1}{12}$ × volume of the wall

$$= \frac{18000000}{12}$$

$$= 1500000 \text{ cm}^3$$

Total volume of the bricks in the wall = volume of the wall − volume of the mortar

$$= (18000000 - 1500000) \text{ cm}^3$$

$$= 16500000 \text{ cm}^3$$

∴ Number of bricks = volume of bricks ÷ volume of one brick

$$= 16500000 \div 2062.5$$

$$= 8000$$

Solution 11:

External length of the box = 36 cm

External breadth of the box = 25 cm

External height of the box = 16.5 cm

∴ External volume of the box = $(36 \times 25 \times 16.5) \text{ cm}^3 = 14850 \text{ cm}^3$

Internal length of the box = $[36 - (1.5 \times 2)]$ cm = 33 cm

Internal breadth of the box = $[25 - (1.5 \times 2)]$ cm = 22 cm

Internal height of the box = $(16.5 - 1.5)$ cm = 15 cm

∴ Internal volume of the box = $(33 \times 22 \times 15) \text{ cm}^3 = 10890 \text{ cm}^3$

Thus, volume of iron used in the box

= External volume of the box − Internal volume of the box

$$= (14850 - 10890) \text{ cm}^3$$

$$= 3960 \text{ cm}^3$$

It is given that 1 cm^3 of iron weighs 15 g.

∴ Weight of the empty box $= 3960 \times 15$

$$= 59400 \text{ g}$$

$$= \frac{59400}{1000} \qquad (\because 1\text{ kg} = 1000\text{ g})$$

$$= 59.4\text{ kg}$$

Solution 12:

Length of the box $= 5$ m

Breadth of the box $= 3$ m

$$\text{Area of the sheet required} = \frac{\text{total cost}}{\text{cost per metre square}}$$

Let h m be the height of the box.

Then area of the sheet = total surface area of the box

$$= 2(lb + lh + bh)\text{ m}^2$$

$$= 2(5 \times 3 + 5 \times h + 3 \times h)\text{ m}^2$$

$$= 2(15 + 8h)$$

$$= (30 + 16h)\text{ m}^2$$

$$\text{Now, } 30 + 16h = \frac{6480}{120}$$

$$\Rightarrow 30 + 16h = 54$$

$$\Rightarrow 16h = 24$$

$$\Rightarrow h = 1.5\text{ m}$$

$\therefore$ The height of the box is 1.5 m.

Solution 13: Length of the cuboid $= 16$ m

Suppose that the breadth and height of the cuboid are 3x m and 2x m, respectively.

Then $1536 = 16 \times 3x \times 2x$

$$\Rightarrow 1536 = 16 \times 6x^2$$

$$\Rightarrow x^2 = \frac{1536}{96}$$

$$\Rightarrow x^2 = 16$$

$$\Rightarrow x = \sqrt{16}$$

$$\Rightarrow x = 4$$

$\therefore$ The breadth and height of the cuboid are 12 m and 8 m, respectively.

Solution 14: Volume of the dining hall $= (20 \times 16 \times 4.5)\ m^3 = 1440\ m^3$

Volume of air required by each person $= 5\ m^3$

$$\therefore \text{ Capacity of the dining hall } = \frac{\text{volume of dining hall}}{\text{volume of air required by each person}}$$

$$= \frac{1440}{5}$$

$$= 288 \text{ persons}$$

Solution 15: Length of the classroom $= 10$ m

Breadth of the classroom $= 6.4$ m

Height of the classroom $= 5$ m

Area of the floor $=$ length $\times$ breadth $= 10 \times 6.4\ m^2$

$$\text{No. of students} = \frac{\text{area of the floor}}{\text{area given to one student on the floor}}$$

$$= \frac{10 \times 6.4}{1.6}$$

$$= \frac{640}{16}$$

$$= 40 \text{ students}$$

Now, volume of the classroom$=10 \times 6.4 \times 5\ m^3$

$$\therefore \text{ Air required by each student} = \frac{\text{volume of the room}}{\text{number of students}}$$

$$= \frac{10 \times 6.4 \times 5}{40}\ m^3$$

$$= 8\ m^3$$

Solution 16:

Length of the cuboid $= 14$ cm

Breadth of the cuboid $= 11$ cm

Let the height of the cuboid be x cm.

Surface area of the cuboid $= 758\ cm^2$

Then,

$758 = 2(14 \times 11 + 14 \times x + 11 \times x)$

$\Rightarrow 758 = 2(154 + 14x + 11x)$

$\Rightarrow 758 = 2(154 + 25x)$

$\Rightarrow 758 = 308 + 50x$

$\Rightarrow 50x = 758 - 308$

$\Rightarrow 50x = 450$

$\Rightarrow x = \frac{450}{50}$

$\Rightarrow x = 9$

∴ The height of the cuboid is 9 cm.

Solution 17:

Volume of the water that falls on the ground = area of ground × depth

$= 20000 \times 0.05 \text{ m}^3$

$= 1000 \text{ m}^3$

Solution 18: Here, a = 9 m

Volume of the cube = a^3

$= 9^3 \text{ m}^3$

$= 729 \text{ m}^3$

Lateral surface area of the cube = $4\ a^2$

$= 4 \times 9^2$

$= 4 \times 81 \text{ m}^2$

$= 324 \text{ m}^2$

Total surface area of the cube = $6a^2 = 6 \times 9^2$

$= 6 \times 81 \text{ m}^2$

$= 486 \text{ m}^2$

∴ Diagonal of the cube = $\sqrt{3}a = \sqrt{3} \times 9 = 15.57\ m^3$

Solution 19: Suppose that the side of cube is $x$cm.

Total surface area of cube $= 1176$ sq cm

Then

$1176 = 6x^2$

$\Rightarrow x^2 = \frac{1176}{6}$

$\Rightarrow x^2 = 196$

$\Rightarrow x = \sqrt{196}$

$\Rightarrow x = 14$

i.e., the side of the cube is 14 cm.

$\therefore$ Volume of the cube $= x^3 = 14^3 = 2744\ cm^3$

Solution 20: Suppose that the side of cube is $x$cm.

Lateral surface area of the cube $= 900$ cm^2

Then

$900 = 4x^2$

$\Rightarrow x^2 = \frac{900}{4}$

$\Rightarrow x^2 = 225$

$\Rightarrow x = \sqrt{225}$

$\Rightarrow x = 15$

i.e., the side of the cube is 15 cm.

$\therefore$ Volume of the given cube $= x^3 = 15^3 = 3375\ cm3$

Solution 21: Suppose that the side of the given cube is $x$cm.

Volume of the cube $= 512$ cm^3

Then

$512 = x^3$

$\Rightarrow x = \sqrt[3]{512}$

$\Rightarrow x = (8^3)^{\frac{1}{3}}$

$\Rightarrow x = 8$

i.e., the side of the cube is 8 cm.

$\therefore$ Surface area of the cube $= 6x^2 = 6 \times 8^2 = 384\ cm^2$

Solution 22: Three cubes of metal with edges 3cm, 4cm and 5 cm are melted to form a single cube.

$\therefore$ Volume of the new cube = sum of the volumes the old cubes

$= (3^3 + 4^3 + 5^3)\ \text{cm}^3$

$= (27 + 64 + 125)\ \text{cm}^3$

$= 216\ \text{cm}^3$

Suppose the edge of the new cube $= x\text{cm}$

Then we have:

$216 = x^3$

$\Rightarrow x = (216)^{\frac{1}{3}}$

$\Rightarrow x = (6^3)^{\frac{1}{3}}$

$\Rightarrow x = 6$

i.e., the edge of the new cube is 6 cm.

$\therefore$ Lateral surface area of the new cube $= 4x^2 = 4 \times 6^2 = 144\ cm^2$

Solution 23: Length of the longest pole = length of the diagonal of the room

$= \sqrt{l^2 + b^2 + h^2}\ \text{m}$

$= \sqrt{10^2 + 10^2 + 5^2}\ \text{m}$

$= \sqrt{100 + 100 + 25}$

$= \sqrt{225}$

$= 15\ \text{m}$

Solution 24: Let the length, breadth and height (or depth) of the cuboid be l cm, b cm and h cm, respectively.

$\therefore l + b + h = 19$ .....(i)

Also,

Length of the diagonal = 11 cm

$\Rightarrow \sqrt{l^2 + b^2 + h^2} = 11$

Squaring on both sides, we get

$l^2 + b^2 + h^2 = 121$ ... (ii)

Squaring (i), we get

$(l + b + h)^2 = 19^2$

$\Rightarrow l^2 + b^2 + h^2 + 2(lb + bh + hl) = 361$

$\Rightarrow 121 + 2(lb + bh + hl) = 361$ [Using (ii)]

$\Rightarrow 2(lb + bh + hl) = 361 - 121 = 240\ cm^2$

Thus, the surface area of the cuboid is 240 cm^2.

Solution 25: Let the initial edge of the cube be a units.

$\therefore$ Initial surface area of the cube $= 6a^2$ square units

New edge of the cube $= a + 50\%$ of a

$$= a + \frac{50}{100}a$$

$$= \frac{100a + 50a}{100}$$

$$= \frac{150a}{100}$$

$$= 1.5a \text{ units}$$

$\therefore$ New surface of the cube $= 6(1.5a)^2 = 13.5a^2$ square units

Increase in surface area of the cube $= 13.5a^2 - 6a^2 = 7.5a^2$ square units

$\therefore$ Percentage increase in the surface area of the cube

$$= \frac{\text{Increase in surface area of the cube}}{\text{Initial surface area of the cube}} \times 100\%$$

$$= \frac{7.5a^2}{6a^2} \times 100\%$$

$$= 125\%$$

Solution 26: Let the length, breadth and height of the cuboid be a, b and c, respectively.

$\therefore$ Surface area of the cuboid, $S = 2(ab + bc + ca)$

Volume of the cuboid, $V = abc$

Now,

$$\frac{S}{V} = \frac{2(ab + bc + ca)}{abc}.$$

$$\Rightarrow \frac{S}{V} = 2\left(\frac{ab}{abc} + \frac{bc}{abc} + \frac{ca}{abc}\right)$$

$$\Rightarrow \frac{S}{V} = 2\left(\frac{1}{c} + \frac{1}{a} + \frac{1}{b}\right)$$

$$\Rightarrow \frac{1}{V} = \frac{2}{S}\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right).$$

Solution 27:

Width of the canal $= 30$ dm $= 3$ m $\quad (1$ m $= 10$ dm$)$

Depth of the canal $= 12$ dm $= 1.2$ m

Speed of the water flow $= 20$ km/h $= 20000$ m/h

$\therefore$ Volume of water flowing out of the canal in 1 h $= 3 \times 1.2 \times 20000 = 72000$ m^3

So, Volume of water flowing out of the canal in 30 minutes $= \frac{72000}{2}$ m^3 $= 36000\ m^3$

Height of standing water on field $= 9$ cm $= 0.09$ m $\quad (1$ m $= 100$ cm$)$

Assume that water flows out of the canal in 30 minutes. Then,

Area of the field irrigated

$$= \frac{\text{Volume of water flowing out of the canal in 30 minutes}}{\text{Height of standing water on the field}}$$

$$= \frac{36000}{0.09}$$

$$= 400000 \text{ m}^2$$

Thus, the area of the field irrigated is 400000 m^2.

Solution 28: Volume of the solid metallic cuboid $= 9\text{m} \times 8\text{m} \times 2\text{m} = 144 \text{ m}^3$

Volume of each solid cube $= (\text{Edge})^3 = (2)^3 = 8 \text{ m}^3$

$$\therefore \text{ Number of cubes formed } = \frac{\text{Volume of the solid metallic cuboid}}{\text{Volume of each solid cube}} = \frac{144}{8} = 18$$

Thus, the number of cubes so formed is 18.